A Finite-Cutoff Hilbert-Space Model

for Prime–Zero Energy Structure

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Abstract

We construct a finite-cutoff Hilbert-space model for the spectral structure associated with primes and nontrivial zeros of the Riemann zeta function. Given a prime cutoff κ and truncated zero ordinates $\{\gamma_1, \ldots, \gamma_N\}$, we define two finite-dimensional real Hilbert spaces H_{str} (over primes) and H_{null} (over zeros), connected by a linear map $\Phi\colon H_{\mathrm{str}} \to H_{\mathrm{null}}$, and the self-adjoint loop operator $T = \Phi^* \circ \Phi$. We establish the algebraic identity

$$\Delta := E_{\mathrm{str}} - E_{\mathrm{spec}} = \sum_p c_p^2 \|a_p\|^2 - \Big\|\sum_p c_p a_p\Big\|^2$$

by direct expansion; numerically, $\Delta > 0$ and $\eta_{\mathrm{orig}} = \Delta / E_{\mathrm{str}} \in [0.598, 0.704]$ for benchmark parameters $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, $\sigma \in [0.1, 0.95]$. The formal core is self-contained in §§ 2–5; results from Papers 1–2 appear only as context. An analytic proof of $\eta_{\mathrm{orig}} > 0$ remains open. *[Geometric interpretation, not used in any proof]:* The model suggests a picture in which $\sigma = \frac{1}{2}$ plays the role of an energy reference point.

Contents

1 Introduction

1.1 Background and Motivation

The present paper is the third in a series studying the curvature of the Riemann zeta function from a geometric and operator-theoretic perspective. A reader may read §§ 2–5 independently of the series context; §§ 1.1–1.2 and §§ 7–10 provide motivation and interpretation but are not needed for the formal results. The paper is intended to be mathematically self-contained and readable without prior knowledge of the earlier papers in the series. Definitions and formal setup are in §2. The proofs of the main formal statements are in §§ 3–5, and are self-contained. Section 6 records open problems. No external project material is needed for §§ 2–5. The geometric interpretations in §§ 7–10 are clearly separated from the formal content.

Paper 1 [7] introduced the function

$$H_\xi(\sigma, t) := \frac{d^2}{d\sigma^2} \log |\xi(\sigma + it)|^2$$

as a second-derivative decomposition of the log-completed zeta function, and established its local positivity structure. A key finding was the divergence $H_{\mathrm{local}}(\frac{1}{2}, \kappa) \sim 8(\log \kappa)^2 \to \infty$ at $\sigma = \frac{1}{2}$, contrasted with convergence for $\sigma > \frac{1}{2}$. The connection to the Li coefficients λ_n, whose positivity is equivalent to RH [5], was identified.

Paper 2 [8] built on this by constructing explicit Schwartz-class test functions $g^* \in \mathcal{S}_{\mathrm{ad}}$ adapted to the adelic zeta integral. These functions realize the Weil explicit formula as an identity $W(g^* * \check{g}^*) = Z(g^*) - H_{\mathrm{local}}(\sigma, \kappa) + O(\varepsilon)$, providing a bridge between local curvature data and the Weil positivity criterion.

The present paper makes the geometric structure underlying both papers explicit. The central objects — two finite-dimensional Hilbert spaces connected by a linear map Φ, and a self-adjoint loop operator — are motivated by structures appearing in Papers 1 and 2 and are here constructed and analyzed directly.

1.2 Imported Results from Papers 1 and 2

This paper builds on the following results established in Papers 1 and 2:

(R1) The truncated local curvature $H_{\mathrm{local}}(\sigma, \kappa) = \psi_1(\sigma) + \sum_{p \le \kappa} V_p(\sigma)$ satisfies $H_{\mathrm{local}}(\frac{1}{2}, \kappa) \sim 8(\log \kappa)^2 \to \infty$, while $H_{\mathrm{local}}(\sigma, \kappa) \to C(\sigma) < \infty$ for all $\sigma > \frac{1}{2}$. *[Paper 1, Lemma 2]*

(R2) An explicit family of admissible test functions g^* satisfies $W(g^* * \check{g}^*) = Z(g^*) - H_{\mathrm{local}}(\sigma, \kappa) + O(\varepsilon)$. *[Paper 2, Theorem 3.1]*

(R3) The renormalized prime weights $c_p^{\mathrm{ren}} = f_p^{1/2} > 0$ satisfy $D = \sum_p (c_p^{\mathrm{ren}})^2 \approx 9.471 < \infty$. *[Paper 2, §4]*

Reader contract: Results R1–R3 are not used in any proof or theorem-level argument in

§§ 2–5. They appear in § 2.3 (Remark 2.3) for motivation only, and formally in § 10.2 for contextual discussion. §§ 2–5 are a finite-dimensional operator model; they do not require acceptance of any RH-related interpretation.

1.3 What This Paper Proves and What It Imports

Content	Status	Where
$H_{\mathrm{str}}, H_{\mathrm{null}}, \Phi, T$ (§§ 2–5)	Proved here	§§ 2–5
Algebraic identity $\Delta = E_{\mathrm{str}} - E_{\mathrm{spec}}$	Proved here	§4
$\eta_{\mathrm{orig}} \in [0.598, 0.704]$	Numerical (this paper)	§5
$H_{\mathrm{local}}(\frac{1}{2}, \kappa) \sim 8(\log \kappa)^2$ (R1)	Imported [7]	§ 2.3, §§ 8–9
$W(\cdot) = Z(g^*) - H_{\mathrm{local}} + O(\varepsilon)$ (R2)	Imported [8]	§ 10.2
$c_p^{\mathrm{ren}} = f_p^{1/2}$, $D \approx 9.471$ (R3)	Imported [8]	§ 10.2
§§ 7–10 geometric model	Interpretive only	Not in proofs

1.4 Connection to Papers 1 and 2

The present paper isolates a finite-cutoff Hilbert-space structure underlying the energy comparisons of Papers 1 and 2, while the full normalization-level identification with Paper 2 remains open (§ 10.2). The renormalized framework $(Z_{\mathrm{ren}}, H_{\mathrm{ren}}, c_p^{\mathrm{ren}})$ of Paper 2 and the η-framework $(E_{\mathrm{str}}, E_{\mathrm{spec}}, c_p)$ of this paper use different normalizations; a precise normalization-level identification remains open (see § 10.2 and Open Problem 6.5).

1.5 Relation to Existing Work

The construction of $T = \Phi^* \circ \Phi$ from prime resonance vectors provides a concrete Hilbert-space realization outside the Hilbert-Pólya setting: T is explicitly constructed from prime-resonance vectors, and its eigenvalues are not the zero ordinates. The adelic viewpoint connects to Connes' noncommutative geometry approach [1], and the test functions of Paper 2 have the structure of adelic Schwartz functions as in Tate's thesis [10]. The Weil positivity criterion [11, 4] provides the outer framework.

1.6 Outline

Section 2 defines the two Hilbert spaces and their inner products. Section 3 constructs the linear map Φ and the loop operator T. Section 4 proves the variance identity and defines η_{orig}. Section 5 collects numerical results. Section 6 states open problems. Sections 7–10 develop the geometric interpretation (all statements in these sections are explicitly marked as geometric model or heuristic).

2 Two Hilbert Spaces

We work throughout with real Hilbert spaces; all vectors and operators have real entries. Complex phasor language appears in § 3.1 and §§ 7–10 as heuristic motivation only; all formal objects ($H_{\text{str}}, H_{\text{null}}, \Phi, T$) are real.

Throughout this paper, $\kappa > 1$ is a fixed real cutoff parameter, $S(\kappa) := \{p \text{ prime} : p \leq \kappa\}$ the set of active primes, and $\{\gamma_k\}_{k=1}^N$ the first N positive imaginary parts of nontrivial zeros of $\zeta(s)$, ordered $0 < \gamma_1 \leq \gamma_2 \leq \cdots$. The Gaussian damping parameter $\varepsilon > 0$ is fixed throughout.

Definition 2.1 (Prime Space H_{str})**.**

$$H_{\text{str}} := \ell^2(S(\kappa))$$

with basis $\{e_p : p \in S(\kappa)\}$*, standard inner product* $\langle a, b\rangle_{\text{str}} = \sum_p a_p b_p$*, and* $\dim H_{\text{str}} = |S(\kappa)| = \pi(\kappa)$*. The* structural energy *of a weight vector* $c \in H_{\text{str}}$ *is* $E_{\text{str}}(c) = \sum_p c_p^2 \|a_p\|^2$ *(defined in §4 after* Φ *is introduced).*

Definition 2.2 (Zero Space H_{null})**.**

$$H_{\text{null}} := \ell^2(\{\gamma_1, \ldots, \gamma_N\})$$

with basis $\{f_k : k = 1, \ldots, N\}$*, inner product* $\langle u, v\rangle_{\text{null}} = \sum_{k=1}^N u_k v_k$ *(unweighted), and* $\dim H_{\text{null}} = N$*.*

Remark (Why $\sigma = \frac{1}{2}$ is a distinguished reference point)**.** [Context only — not used in any argument in §§ 3–5.]

Two independent proven reasons support this choice:

(i) Symmetry *[proven]:* $\sigma = \frac{1}{2}$ *is the unique fixed point of* $s \mapsto 1 - s$*, the symmetry of the completed zeta function* $\xi(s) = \xi(1 - s)$*.*

(ii) Curvature singularity *[proven, Papers 1–2]:* $H_{\text{local}}(\frac{1}{2}, \kappa) \sim 8(\log \kappa)^2 \to \infty$ *as* $\kappa \to \infty$*, while* $H_{\text{local}}(\sigma, \kappa) \to C(\sigma) < \infty$ *for all* $\sigma > \frac{1}{2}$*.*

$\sigma = \frac{1}{2}$ *is thus a distinguished reference point by symmetry and curvature divergence. Energy near-equality (*$E_{\text{str}} \approx E_{\text{spec}}$*) is NOT observed in the finite-cutoff model (see §5 and §8).*

3 The Linear Map Φ and the Loop Operator T

Convention (Option B throughout): Damping lives in the vectors: $a_p = (e^{-\varepsilon^2\gamma_k^2/2} \sin(\gamma_k \log p))_k$. The H_{null} inner product is unweighted: $\langle u, v\rangle = \sum_k u_k v_k$.

3.1 Motivation for the resonance vectors

[Heuristic motivation — the formal definition is Definition 3.2 below.]

The prime resonance vector at zero ordinate γ_k and prime p arises as the imaginary part of the prime phasor $\varphi_p(u) = e^{iu \log p}$ evaluated at $u = \gamma_k$, with Gaussian regularization for convergence:

$$(a_p)_k = e^{-\varepsilon^2 \gamma_k^2/2} \cdot \sin(\gamma_k \log p).$$

The k-th component measures how strongly prime p resonates with zero ordinate γ_k.

Definition 3.1 (Prime Resonance Vectors)**.** *For each prime $p \in S(\kappa)$, the* canonical resonance vector *at $\sigma = \frac{1}{2}$ is*

$$a_p^{\mathrm{can}} := a_p := \left(e^{-\varepsilon^2 \gamma_k^2/2} \sin(\gamma_k \log p)\right)_{k=1}^{N} \in H_{\mathrm{null}}.$$

At general σ, the resonance vectors depend on σ via

$$a_p^{(\sigma)} := p^{-(\sigma-1/2)} \cdot a_p^{\mathrm{can}}.$$

The vectors $a_p = a_p^{\mathrm{can}} = a_p^{(1/2)}$ at $\sigma = \frac{1}{2}$ are the canonical representatives used in all primary definitions.

Definition 3.2 (Linear Map Φ)**.**

$$\Phi \colon H_{\mathrm{str}} \to H_{\mathrm{null}}, \qquad \Phi(e_p) := a_p^{\mathrm{can}}.$$

Extended linearly: $\Phi(c) = \sum_p c_p a_p^{\mathrm{can}}$ for any $c = \sum_p c_p e_p \in H_{\mathrm{str}}$.

We define Φ as a linear map; no injectivity claim is made for the finite-cutoff model.

Definition 3.3 (Gram Matrices)**.** *The* unnormalized Gram matrix *(canonical) is*

$$G_{pq}^{\mathrm{un}} := G_{pq}^{\mathrm{un,can}} := \langle a_p^{\mathrm{can}}, a_q^{\mathrm{can}} \rangle_{H_{\mathrm{null}}} = \sum_{k=1}^{N} e^{-\varepsilon^2 \gamma_k^2} \sin(\gamma_k \log p) \sin(\gamma_k \log q).$$

The σ-dependent extension is $G^{\mathrm{un}}(\sigma)_{pq} = p^{-(\sigma-1/2)} q^{-(\sigma-1/2)} G_{pq}^{\mathrm{un}}$. The normalized Gram matrix *is*

$$G_{pq}^{\mathrm{norm}} := \frac{G_{pq}^{\mathrm{un}}}{\|a_p^{\mathrm{can}}\| \cdot \|a_q^{\mathrm{can}}\|} = \left\langle \frac{a_p^{\mathrm{can}}}{\|a_p^{\mathrm{can}}\|}, \frac{a_q^{\mathrm{can}}}{\|a_q^{\mathrm{can}}\|} \right\rangle_{H_{\mathrm{null}}}.$$

The matrix G^{norm} is σ-invariant (Proposition 3.5(iv)).

Definition 3.4 (Loop Operator T)**.**

$$T := \Phi^* \circ \Phi \colon H_{\mathrm{str}} \longrightarrow H_{\mathrm{str}}.$$

The matrix representation of T in the prime basis is $T_{pq} = G_{pq}^{\mathrm{un}}$. T is thus identical to the canonical Gram matrix G^{un}, and is explicitly constructed from prime resonance vectors.

Proposition 3.5 (Properties of T)**.** (i) Self-adjoint: $T^* = (\Phi^*\Phi)^* = \Phi^*\Phi = T$.

(ii) Positive semi-definite: $\langle Tc, c \rangle = \|\Phi c\|^2 \geq 0$ *for all $c \in H_{\mathrm{str}}$.*

(iii) Non-negative eigenvalues: *All eigenvalues $\lambda_j \geq 0$.*

(iv) σ-invariance of G^{norm} *[by direct calculation]: For the σ-dependent family $a_p^{(\sigma)}$, the normalized Gram matrix G_{pq}^{norm} is independent of σ.* Proof: *The scalar factor $p^{-(\sigma-1/2)}$ cancels in the normalized ratio.* □

Remark (Spectral structure of T). *The eigenvalues $\lambda_j \geq 0$ of $T = \Phi^* \circ \Phi$ are the squares of the singular values $\sigma_j(\Phi)$: $\lambda_j = \sigma_j(\Phi)^2$. The eigenvalues λ_j are built from the zero ordinates γ_k appearing in the definition of a_p^{can}, but are* not *directly the zeros γ_k themselves. This distinguishes T from the classical Hilbert-Pólya ansatz: T is explicitly constructed, not postulated.*

4 The Variance Identity and η_{orig}

Convention: Objects without σ-argument refer to the canonical ($\sigma = \frac{1}{2}$) version throughout §§ 4–5.

Definition 4.1 (Canonical and σ-dependent Energies). *For a weight vector $c \in H_{\mathrm{str}}$ with components c_p:*

Canonical quantities *(at $\sigma = \frac{1}{2}$):*

$$D^{\mathrm{can}} := \mathrm{diag}\big(\|a_p^{\mathrm{can}}\|^2\big)_{p\in S(\kappa)}, \qquad E_{\mathrm{str}}^{\mathrm{can}} := c^\top D^{\mathrm{can}} c = \sum_p c_p^2 \|a_p^{\mathrm{can}}\|^2,$$

$$E_{\mathrm{spec}}^{\mathrm{can}} := \|\Phi c\|^2 = c^\top G^{\mathrm{un}} c.$$

σ-dependent extensions:

$$D(\sigma)_{pp} := p^{-2(\sigma-1/2)} \cdot D_{pp}^{\mathrm{can}}, \qquad E_{\mathrm{str}}(\sigma) := \sum_p c_p^2\, p^{-2(\sigma-1/2)} \|a_p^{\mathrm{can}}\|^2,$$

$$E_{\mathrm{spec}}(\sigma) := c^\top G^{\mathrm{un}}(\sigma)\, c.$$

The canonical weight vector *in the η-context is*

$$c_p = \sqrt{8} \cdot \frac{\log p}{p}, \qquad p \in S(\kappa).$$

Assumption 4.2 (Positive definiteness of D). *Throughout this paper we assume $D_{pp} = \|a_p\|^2 > 0$ for all $p \in S(\kappa)$. This holds in all numerical experiments conducted here.*

Theorem 4.3 (Algebraic Identity). *The following equality holds by direct expansion:*

$$\Delta := E_{\mathrm{str}} - E_{\mathrm{spec}} = \sum_p c_p^2 \|a_p\|^2 - \Big\|\sum_p c_p a_p\Big\|^2.$$

Proof. Expand $E_{\mathrm{spec}} = \|\Phi c\|^2 = \big\langle \sum_p c_p a_p, \sum_q c_q a_q \big\rangle = \sum_{p,q} c_p c_q \langle a_p, a_q\rangle = c^\top G^{\mathrm{un}} c$. Then $\Delta = c^\top D^{\mathrm{can}} c - c^\top G^{\mathrm{un}} c$, which is the stated identity. □

Observation 4.4 (Non-negativity, numerical)**.**

$$\Delta \geq 0 \quad \textit{equivalently} \quad \eta_{\text{orig}} \geq 0$$

holds for all tested parameters with canonical weights $c_p = \sqrt{8}\log(p)/p$ *(Definition 4.1).* [Numerical only — no unconditional proof for general c.]

Note: $\eta_{\text{orig}} \geq 0$ *for general weight vector* c *is equivalent to* $\lambda_{\max}(D^{-1/2}TD^{-1/2}) \leq 1$*, which is the content of Open Problem 6.2. Numerical experiments show* $\lambda_{\max}(D^{-1/2}TD^{-1/2}) \approx 0.39 \cdot \pi(\kappa) \to \infty$*, so the universal bound* $\lambda_{\max} \leq 1$ *does not hold in general.*

Cauchy-Schwarz does **not** guarantee $\eta_{\text{orig}} \geq 0$*: the correct identity is* $\Delta = c^\top(D^{\text{can}} - G^{\text{un}})c$*; the sign of* Δ *is not determined by the algebraic identity alone.*

Definition 4.5 (η_{orig})**.**

$$\eta_{\text{orig}} := 1 - \frac{E_{\text{spec}}}{E_{\text{str}}} = 1 - \frac{\|\Phi c\|^2}{E_{\text{str}}}.$$

Boundary cases: $\eta_{\text{orig}} = 0$ *iff* $E_{\text{spec}} = E_{\text{str}}$*;* $\eta_{\text{orig}} = 1$ *iff* $\Phi c = 0$.

Remark (σ-dependence via rescaled weights)**.** *The entire* σ*-dependence of* $\eta_{\text{orig}}(\sigma)$ *is carried by the rescaled weight vector* $\tilde{c}_p(\sigma) := c_p \cdot p^{-(\sigma-1/2)}$*:*

$$\eta_{\text{orig}}(\sigma) = 1 - \frac{\tilde{c}(\sigma)^\top G^{\text{un}}\, \tilde{c}(\sigma)}{\tilde{c}(\sigma)^\top D^{\text{can}}\, \tilde{c}(\sigma)}.$$

Proof: *Since* $G^{\text{un}}(\sigma)_{pq} = p^{-(\sigma-1/2)}q^{-(\sigma-1/2)}G^{\text{un}}_{pq}$*, one has* $c^\top G^{\text{un}}(\sigma)c = \tilde{c}^\top G^{\text{un}}\tilde{c}$ *and* $E_{\text{str}}(\sigma) = \tilde{c}^\top D^{\text{can}}\tilde{c}$*.* □

5 Numerical Results

Setup. Zeros γ_k computed via `mpmath.zetazero()`, $N = 100$ throughout. σ grid: $\{0.1, 0.2, \ldots, 0.9, 0.95\}$. Eigenvectors and eigenvalues: NumPy `linalg.eigh()`. Requirements: mpmath ≥ 1.3; NumPy ≥ 1.24. All computations via `eta_verification.py`; companion code at `https://github.com/utehrani/analysislab-nt`. Algebraic identity (Theorem 4.3) numerically verified to rounding error $< 10^{-14}$.

All results in this section are numerical. Parameters tested: $\kappa \in \{23, 53, 101, 199\}$*,* $\varepsilon \in \{0.02, 0.05, 0.10\}$*,* $\sigma \in [0.1, 0.95]$*,* $N = 100$*.*

Observation 5.1 (η_{orig} positive across all tested parameters)**.** $\eta_{\text{orig}} > 0$ *in all tested parameter combinations. For the benchmark slice* $(\kappa, \varepsilon, N) = (53, 0.05, 100)$*,* $\sigma \in [0.1, 0.95]$*:*

$$\eta_{\text{orig}} \in [0.598,\ 0.704].$$

The function $\sigma \mapsto \eta_{\text{orig}}(\sigma)$ *is strictly monotone decreasing. Note: the range* $[0.598, 0.704]$ *is strongly* ε*-dependent; no universal range over all* (κ, ε) *is claimed.*

Table 5.2: $\eta_{\text{orig}}(\sigma)$ for $(\kappa, \varepsilon, N) = (53, 0.05, 100)$. [Numerical]

σ	η_{orig}
0.10	0.70441977
0.20	0.69826865
0.30	0.69038398
0.40	0.68072993
0.50	0.66926873
0.60	0.65602069
0.70	0.64110102
0.80	0.62472010
0.90	0.60715692
0.95	0.59802896

Observation 5.2 (Uniform spectral bound $B_{\max} \leq 0.106 < 1$)**.** *Define the mode contribution $B_j(\sigma) := \lambda_j(\sigma)|\langle c, u_j(\sigma)\rangle|^2/E_{\text{str}}(\sigma)$ with consistency identity $\sum_j B_j(\sigma) = 1 - \eta_{\text{orig}}(\sigma)$ [algebraic]. The empirical diagnostic $B_{\max}(\sigma) := \max_{j:\lambda_j>1} B_j(\sigma)$ satisfies*

$$B_{\max}(\kappa, \varepsilon, \sigma) \leq 0.106 < 1$$

for all tested $(\kappa, \varepsilon, \sigma)$. Selected values:

κ	$\varepsilon = 0.02$	$\varepsilon = 0.05$	$\varepsilon = 0.10$
23	*0.1047*	*0.0220*	*0.0000*
53	*0.0793*	*0.0737*	*0.0000*
101	*0.0537*	*0.0844*	*0.0324*
199	*0.0891*	*0.0461*	*0.1058*

Observation 5.3 ($B_{\max}(\sigma)$ numerically σ-invariant)**.** *For $(\kappa, \varepsilon) = (53, 0.05)$, $B_{\max}(\sigma) = 0.0737$ for all tested $\sigma \in \{0.3, 0.4, 0.5, 0.6, 0.7\}$. Analytical explanation: open (Open Problem 6.4).*

Remark (Cancellation mechanism, numerical analysis)**.** *The mechanism underlying $B_{\max} < 1$ is a cancellation between the monotone weight vector $c_p = \sqrt{8}\log(p)/p$ and the oscillatory eigenvectors u_j of G^{un}: for $\kappa = 53$, eigenvectors $u_1, \ldots, u_4$ each exhibit 10 sign changes along the prime ordering. Cancellation rates 67–97% observed for $\kappa \in \{23, 53, 101, 199\}$. Formal proof: Open Problem 6.3.*

6 Open Problems

(a) Within the η-framework of this paper:

Open Problem (Analytic positivity)**.** *Prove analytically that $\eta_{\text{orig}} > 0$ for the canonical weight vector $c_p = \sqrt{8}\log(p)/p$. Numerically confirmed: $\eta_{\text{orig}} \in [0.598, 0.704]$ for $(\kappa, \varepsilon, N) = (53, 0.05, 100)$, $\sigma \in [0.1, 0.95]$ (Table 5.2).*

Open Problem (Spectral bound for canonical weights)**.** *Let* $T_{\mathrm{ren}} := D^{-1/2}TD^{-1/2}$*. Numerical experiments show* $\lambda_{\max}(T_{\mathrm{ren}}) \approx 0.39 \cdot \pi(\kappa) \to \infty$*; the universal bound* $\lambda_{\max} < 1$ *(for all* c*) does not hold. However, for the canonical weight vector* $c_p = \sqrt{8}\log(p)/p$*, the quantity* $\langle T_{\mathrm{ren}}\tilde{c}, \tilde{c}\rangle$ *satisfies* $\langle T_{\mathrm{ren}}\tilde{c}, \tilde{c}\rangle < 1$ *(equivalently* $\eta_{\mathrm{orig}} > 0$*) for all tested* $\kappa \in \{23, 53, 101, 199, 503, 1009\}$*. Prove this analytically. The canonical* c_p *appears orthogonal to the dominant eigenmode of* T_{ren} *(localized at* $\gamma_1 = 14.13\ldots$*); proving this orthogonality is the core open problem.*

Open Problem (Analytical cancellation bound)**.** *Prove analytically:*

$$\max_{j\colon \lambda_j > 1} \frac{|\langle c, u_j\rangle|^2}{\|c\|^2} \le C_1 \ll 1 \quad \textit{uniformly in } \kappa, j.$$

Approach: Abel summation for $c_p \sim (\log p)/p$ *against oscillatory partial sums* $\sum_{q \le p}(u_j)_q$*, without reference to RH. Numerically:* $C_1 < 0.36$ *for* $(\kappa, \varepsilon) = (53, 0.05)$*.*

Open Problem (σ-invariance of $B_{\max}$ analytically)**.** *The numerical* σ*-invariance of* $B_{\max}(\sigma)$ *(consistent with Proposition 3.5(iv)) is not yet derived analytically from the weight rescaling* $c \mapsto \tilde{c}(\sigma)$*.*

(b) Beyond the scope of this paper:

Open Problem (Renormalized shell energy)**.** *The renormalized shell energy* $H_{\mathrm{ren}}(\kappa, \varepsilon(\kappa))$ *for canonical coupling* $N_{\mathrm{eff}} = \pi(\kappa)$ *is not yet computed. This lies in the Paper 2 normalization framework (see § 10.2).*

Remark (Connection to the Riemann Hypothesis)**.** [Geometric interpretation — not a theorem.] *In the geometric model, RH is modeled heuristically as injectivity of an idealized linear map* Φ*. No formal connection between injectivity of the finite-cutoff* Φ *and* $\eta_{\mathrm{orig}} > 0$ *is established here. The quantitative convexity inequality of Papers 1–2 is an open problem in the earlier framework and is* ***not*** *addressed in this paper.*

7 Four Interpretive Levels (Geometric Interpretation)

Notation for §§ 7–10. Every statement using model labels (*prime resonance, interference field, resonance frequencies, energy reference point*) is a geometric interpretation — not a theorem. Formal objects retain their definitions from §§ 2–5. No map ($\Phi_{01}, \Phi_{12}, \Phi_{23}$), no intermediate space $\mathcal{I}$, and no duality statement in §§ 7–10 is part of the theorem-level content.

[**Model 7.1**] (Four Interpretive Levels). We organize the arithmetic and spectral structures into four levels:

$\Omega_0 : \mathbb{N}$ (arithmetic, discrete)

$\Omega_1 : H_{\mathrm{str}}$ (prime exponent lattice, structural energy)

$\Omega_2 : \mathcal{I}$ (interference fields)

$\Omega_3 : H_{\mathrm{null}}$ (spectral space, zero resonance frequencies) with model maps (mnemonics only, not formal objects): $\Phi_{01} : \mathbb{N} \to H_{\mathrm{str}}$, $\Phi_{12} : H_{\mathrm{str}} \to \mathcal{I}$, $\Phi_{23} : \mathcal{I} \to H_{\mathrm{null}}$.

[**Heuristic 7.3**] (Zeros as Resonance Frequencies). In this geometric picture, the zeros γ_k may be viewed as resonance frequencies of the prime interference field: the condition $\zeta(\sigma + i\gamma_k) = 0$ is modeled as destructive interference of the prime resonance waves $p^{-\sigma - i\gamma_k}$ at frequency γ_k. *This is a heuristic model picture.*

8 The Energy Plane

[**Model 8.1**] (Energy Coordinates). For canonical weight vector c and parameter σ, define $X(\sigma) := \sqrt{E_{\mathrm{str}}(\sigma)}$ and $Y(\sigma) := \sqrt{E_{\mathrm{spec}}(\sigma)}$.

[**Numerically verified,** $(\kappa, \varepsilon, \sigma) = (53, 0.05, 0.5)$]**:** Energy near-equality does NOT hold in the finite-cutoff model:

$$E_{\mathrm{spec}}/E_{\mathrm{str}} \approx 0.331 \qquad (\text{not } \approx 1).$$

$\sigma = \frac{1}{2}$ remains distinguished by proven properties (symmetry and curvature divergence), NOT by energy equilibrium.

[**Model 8.3**] (Variance Identity in the Energy Plane). $\Delta = E_{\mathrm{str}} - E_{\mathrm{spec}} = X^2 - Y^2 = (X + Y)(X - Y)$. The condition $\eta_{\mathrm{orig}} > 0$ is represented as $X > Y$, observed numerically throughout (Observation 4.4).

9 Why $\sigma = \frac{1}{2}$ is Special: Four Reasons

Reason (i) is proven. Reason (ii) is numerical. Reason (iii) is proven. Reason (iv) is a conceptual model only.

(i) Symmetry [proven]: $\sigma = \frac{1}{2}$ is the unique fixed point of $s \mapsto 1 - s$, the symmetry underlying $\xi(s) = \xi(1 - s)$.

(ii) Energy dominance [numerical, §5]: $E_{\mathrm{str}}(\frac{1}{2}) > E_{\mathrm{spec}}(\frac{1}{2})$ for all tested (κ, ε), with $E_{\mathrm{spec}}/E_{\mathrm{str}} \approx 0.331$ at $(\kappa, \varepsilon) = (53, 0.05)$. For the benchmark slice, $\eta_{\mathrm{orig}}(\sigma)$ is strictly monotone decreasing on $[0.1, 0.95]$ (Table 5.2). Energy near-equality ($E_{\mathrm{str}} \approx E_{\mathrm{spec}}$) is NOT observed.

(iii) Curvature singularity [proven, Papers 1–2]:

$$H_{\mathrm{local}}(\tfrac{1}{2}, \kappa) \sim 8(\log \kappa)^2 \to \infty, \qquad H_{\mathrm{local}}(\sigma, \kappa) \to C(\sigma) < \infty \text{ for } \sigma > \tfrac{1}{2}.$$

(iv) [Conceptual model, NOT numerically verified]: In the energy-plane model of §8, one might expect $\sigma = \frac{1}{2}$ to be the "45°-point" where $X(\sigma) \approx Y(\sigma)$ in an idealized infinite model. This is not confirmed in the finite-cutoff numerics: $\sqrt{E_{\mathrm{spec}}/E_{\mathrm{str}}} \approx 0.575 \neq 1$ at $(\kappa, \varepsilon) = (53, 0.05)$.

10 Duality $\mathbb{N} \leftrightarrow \mathbb{C}$ and Outlook

[**Model 10.1**] (Structural Duality).

Structural side Ω_1	**Spectral side Ω_3**
Basis: primes p	Basis: zero ordinates γ_k
Energy: E_{str}	Energy: E_{spec}
Structure: multiplication	Structure: interference
Tool: Euler product	Tool: Explicit formula

[**Model 10.2**] (Connection to Weil Positivity). *[Contextual only — not part of the formal proof core §§ 2–6.]*

In Paper 2, the Weil functional $W(g^* * \check{g}^*)$, whose positivity for all admissible g^* is equivalent to RH [11, 4], satisfies $W(g^* * \check{g}^*) = Z(g^*) - H_{\mathrm{local}}(\sigma, \kappa) + O(\varepsilon)$. This suggests an energy-balance analogy with the η-framework, but the η-framework and Weil framework live in different normalizations. No theorem in this paper identifies $(E_{\mathrm{str}}, E_{\mathrm{spec}}, \Delta)$ with $(Z_{\mathrm{ren}}, H_{\mathrm{ren}}, Z_{\mathrm{off}})$ (see Open Problem 6.5).

Outlook 10.3. Direct next problems: (1) analytic proof of $\eta_{\mathrm{orig}} > 0$ via uniform partial summation (Open Problem 6.3); (2) spectral bound for canonical weights (Open Problem 6.2); (3) renormalized shell energy $H_{\mathrm{ren}}(\kappa, \varepsilon(\kappa))$ (Open Problem 6.5). Broader questions include the connection to Connes' program [1] and whether the test function family $\{g^*_{\sigma,\varepsilon}\}$ is a complete witness system for the Weil criterion [4].

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A Variable Reference

Symbol	Definition	Status
κ	prime cutoff	parameter
$S(\kappa)$	primes $p \le \kappa$	definition
γ_k	k-th zero ordinate (positive imag. part)	input
ε	Gaussian damping	parameter
$a_p^{\text{can}} = a_p$	canonical resonance vector at $\sigma = \frac{1}{2}$	§3.1
$a_p^{(\sigma)}$	$= p^{-(\sigma-1/2)} a_p^{\text{can}}$	§3.1
G_{pq}^{un}	canonical unnormalized Gram matrix	§3.3
$G^{\text{un}}(\sigma)_{pq}$	σ-dependent Gram	§3.3
G_{pq}^{norm}	normalized Gram; σ-invariant [proven]	§3.3
$T = \Phi^*\Phi$	loop operator $= G^{\text{un}}$	§3.4
D^{can}	$\text{diag}(\Vert a_p^{\text{can}}\Vert^2)$	§4.1
$E_{\text{str}}^{\text{can}}$	$= c^\top D^{\text{can}} c$	§4.1
$E_{\text{spec}}^{\text{can}}$	$= \Vert\Phi c\Vert^2 = c^\top G^{\text{un}} c$	§4.1
Δ	$= E_{\text{str}} - E_{\text{spec}}$ [algebraic]; sign numerical	§4.2a/b
η_{orig}	$= \Delta/E_{\text{str}}$	§4.3
$\tilde{c}_p(\sigma)$	$= c_p \cdot p^{-(\sigma-1/2)}$	§4.4
c_p	$= \sqrt{8}\log(p)/p$ (canonical weight)	§4.1
$B_j(\sigma)$	$= \lambda_j \vert\langle c, u_j\rangle\vert^2 / E_{\text{str}}$; $\sum_j B_j = 1 - \eta_{\text{orig}}$	§5.1
$B_{\max}(\sigma)$	$= \max_{j:\lambda_j>1} B_j$ [numerical threshold]	§5.1

B Non-Circularity Remarks

This paper uses no circular reasoning with respect to RH. The construction of $T = \Phi^*\Phi$ from prime resonance vectors is explicit and postulates neither GUE statistics, Montgomery's pair correlation, nor the Hilbert-Pólya conjecture. Geometric model content (§§ 7–10) is always marked as such and is not used in any formal argument.

The following distinctions are maintained throughout:

- The identity $\Delta = E_{\text{str}} - E_{\text{spec}}$ is an algebraic theorem (Theorem 4.3); the sign $\Delta \geq 0$ is numerical only — these are distinct levels.
- The eigenvalues λ_j of $T = \Phi^*\Phi$ are the squares of the singular values of Φ, not the zero ordinates γ_k themselves.
- No claim relating η_{orig} to RH is proved or used.

C Sample Output of `eta_verification.py`

The following output was produced with $\kappa = 53$, $\varepsilon = 0.05$, $N = 100$, $\sigma = 0.5$:

```
-- NUM4 Benchmark (kappa=53, eps=0.05, sigma=0.5, N=100) --
eta_orig = 0.66926873
Reference:  0.66926893   (Delta = 1.97e-07)

-- NUM2 Normative table (kappa=53, eps=0.05, N=100) --------
 sigma | eta_orig
 0.10  | 0.70441977
 0.20  | 0.69826865
 0.30  | 0.69038398
 0.40  | 0.68072993
 0.50  | 0.66926873
 0.60  | 0.65602069
 0.70  | 0.64110102
 0.80  | 0.62472010
 0.90  | 0.60715692
 0.95  | 0.59802896

Monotone decreasing? True
```

Requirements: Python 3.x, NumPy $\geq$ 1.24, mpmath $\geq$ 1.3. Command: `python eta_verification.py`

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